Common setup (run once)

```
pip install nltk
```

```
import nltk

# Download the NLTK datasets/models needed by the lab programs.
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('brown')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('universal_tagset')
nltk.download('conll2000')
nltk.download('maxent_ne_chunker_tab')
nltk.download('words')
```

Run each program in Jupyter Notebook, Google Colab, or VS Code with Python installed. Some programs download corpora on first run and need an internet connection.

6. Implementation of stemming and lemmatization

Aim: To reduce words to their stems and dictionary base forms.

```
import nltk
from nltk.stem import PorterStemmer, WordNetLemmatizer

# Download WordNet, which the lemmatizer uses.
nltk.download('wordnet')
nltk.download('omw-1.4')

# Initialize the stemmer and lemmatizer.
stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

# Words for demonstrating stemming and lemmatization.
words = ['playing', 'played', 'studies', 'cars', 'running']

print(f"{'Word':<12}{'Stem':<12}{'Lemma':<12}")
print("-" * 36)

for word in words:
    # Stemming removes word endings using heuristic rules.
    stem = stemmer.stem(word)

    # Lemmatization finds a dictionary base form.
    # The default POS is noun, so specify verb for these examples.
    lemma = lemmatizer.lemmatize(word, pos='v')

    print(f"{word:<12}{stem:<12}{lemma:<12}")
```

Sample output

```
Word        Stem        Lemma
------------------------------------
playing     play        play
played      play        play
studies     studi       study
cars        car         cars
running     run         run
```

Explanation:

• **Stemming:** Removes or modifies word endings without necessarily producing a valid dictionary word. For example, `studies → studi`.

• **Lemmatization:** Uses vocabulary and morphological rules to find a base form. For example, `studies → study` when treated as a verb.

• Lemmatization depends on the correct part-of-speech (POS) input. Here, `cars` remains `cars` because it is interpreted as a verb by the chosen call, rather than as a plural noun.